Robot Perception, Localization and Navigation

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Neoscholar - 2025

Robot

Sensing and Perception

Outline

1. Robot Sensing
2. Principles of Sensors
   - IMU, GPS
   - LiDAR, Radar, Cameras
3. Sensor Models

Disclaimer: These slides can include material from different sources.
We would like to thank the colleagues who shared the resources used in these slides.

Robot Sensing

- Similar to the human nervous system.
- The five senses often said: for the outside world.
- The remaining three senses: for the robot itself.

Robot Sensing

Robot sensing relies on sensors:

Sensors

- By definition, a sensor is a device that produces an output signal for the purpose of sensing of a physical phenomenon.
- By using sensors, a robot can perceive the environment, covering not only its exterior, such as the distance to an object, but also its own components (i.e. interior), such as its movement.

Sensing Usage

- ***Proprioceptive sensors:*** measure values **internally to the** robot.
- Examples: motor speed, wheel load, heading of the robot, battery status

- ***Exteroceptive sensors:*** gather information **from the robot environment**,
- Examples: distance to objects, object color, 3D map, intensity of the ambient light

Sensors

Sensing Principle

- ***Passive sensors:*** Measure energy coming from the environment

  * Cameras: intensity-based
  * Tactile sensors: intensity-based
  * Motor sensors (brush encoders): electricity-shift

- ***Active sensors:*** Emit their proper energy and measure the reaction, potentially more effective but depends on the characteristics of the environment

  * Laser range finder: Time-of-Flight (ToF)
  * Ultrasound: ToF
  * Radar: ToF
  * Infrared: phase-shift

Characterization of Sensor Performance

- **Dynamic Range:** measure the ratio between the maximum and the minimum input values that can be measured by the sensor.
- **Resolution:** minimum difference that can be measured between two values (for digital sensors it is usually related to the A/D conversion)
- **Response:** variation of output signal as function of the input signal, better when is linear
- **Bandwidth or Frequency:** the (max) speed with which a sensor can provide a stream of readings, one has also to consider phase (delay) of the signal

Characterization of Sensor Performance

- **Sensitivity**: ratio of output change to input change, dy/dx (e.g., magnitude of change of the output of a visual sensor in relation to a change in the illumination)

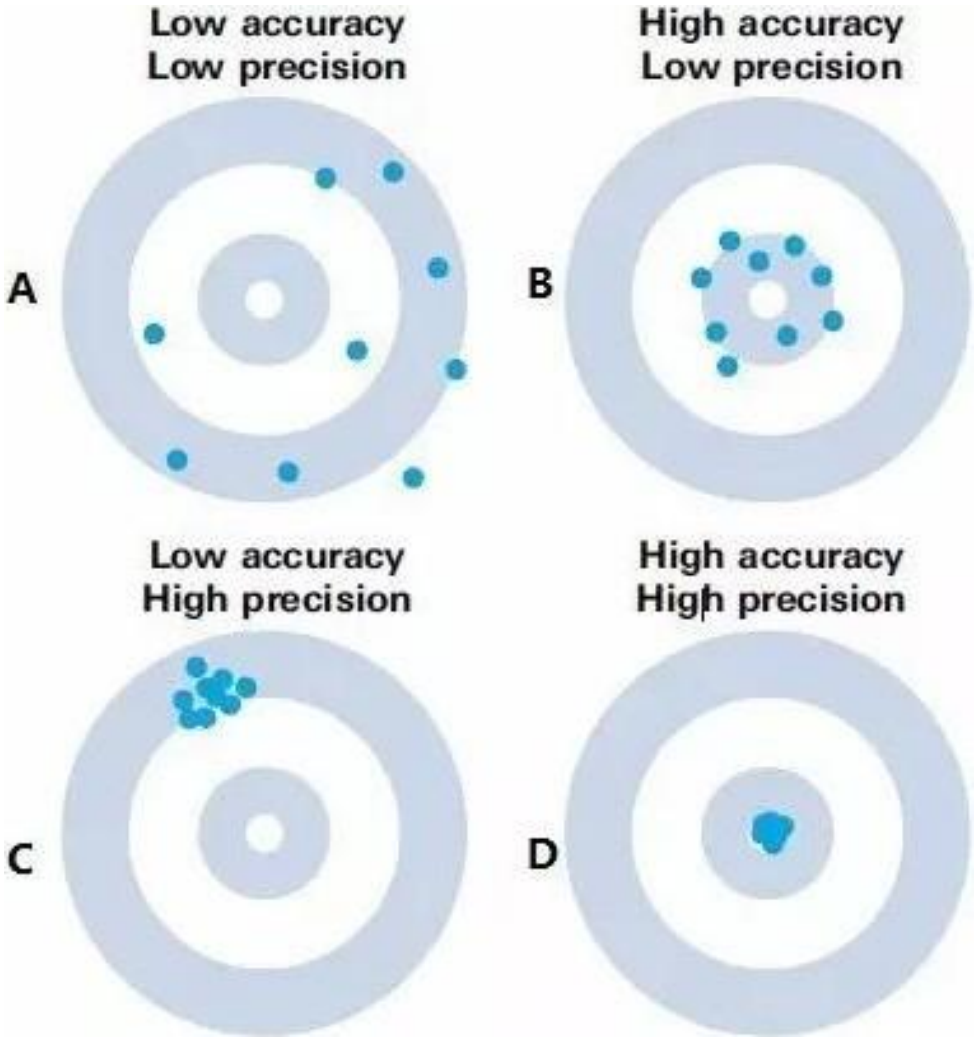
- **Cross-sensitivity** (and cross-talk): sensitivity to (other) environmental parameters (e.g. temperature, magnetic field caused by ferrous materials) and/or influence exert by other active sensors. In general, sensor sensitivity is negatively affected by cross-sensitivity.

- **Error / Accuracy**: deviation between sensor's output and the true value

Since the true value (the "ground truth") can be hard to assess, establishing a confident characterization of sensor sensitivity can be difficult in practice

Sensors

Accuracy vs. Precision



Sensors

What do we get from sensors?

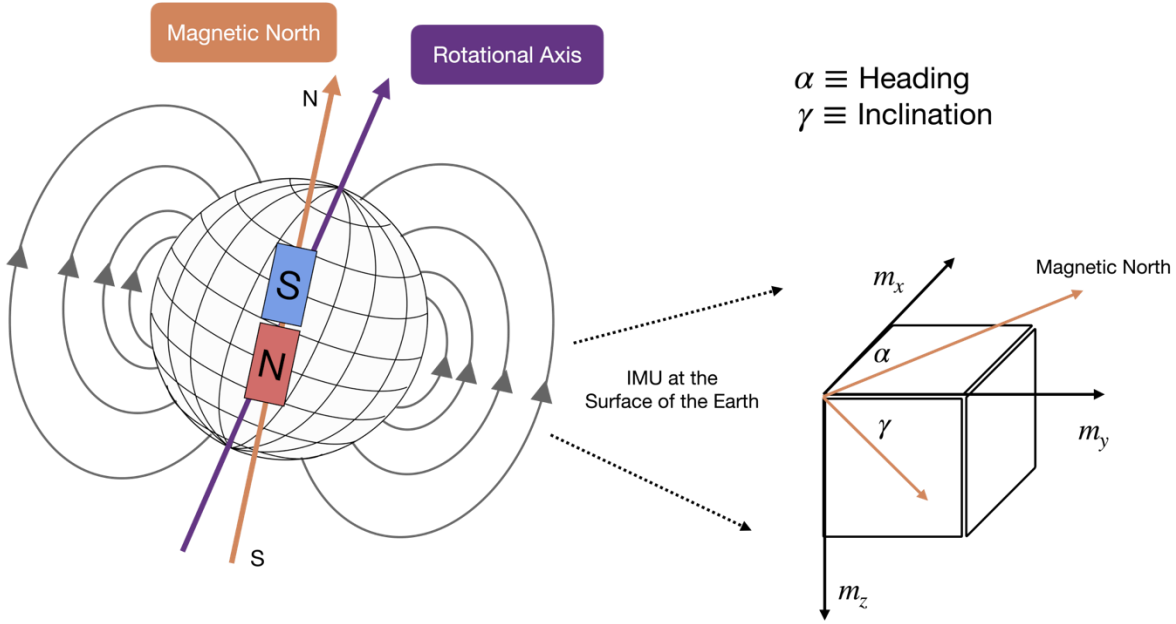
rotary encoder	—>	angle
IMU	—>	acceleration and angular speed
thermometer	—>	temperature
potentiometer	—>	position
accelerometer	—>	acceleration
camera	—>	image
touch sensor	—>	contact
compass	—>	magnetism
lidar	—>	distance
sonar	—>	distance
radar	—>	distance and velocity
GPS	—>	geolocation

Dead Reckoning Sensors

- **Odometry sensors:** Motor Encoders to measure wheel rotation (already covered in last lecture)
- **Heading / orientation sensors:** Compass / Magnetometers
- **Inertial sensors** (measure forces, non-inertial effects): Gyroscope, Accelerometer

Proprioceptive Sensors

Compass / Magnetometers



Magnetic field on earth

- absolute measure for orientation (even birds use it for migrations)

Variety of solutions to measure magnetic/true north

- mechanical magnetic compass
- Gyrocompass
- direct measure of the magnetic field (Hall-effect, magneto-resistive sensors)

The sensor typically provides **three-axis measurements**:

- Mx, My, Mz → Magnetic field strength along X, Y, Z axes, respectively (in microtesla, μT).
- raw readings from the **Earth's magnetic field plus any local magnetic interference** (like nearby metals or electronics).

Using Mx, My, Mz, **Heading** (Azimuth) can be calculated:

1. Angle between magnetic north and the sensor's forward direction (usually in degrees).
2. Typically calculated in 2D using:

$$\text{Heading} = \arctan2(\text{My},\text{Mx})$$

which gives compass direction (0° = North, 90° = East, etc.)

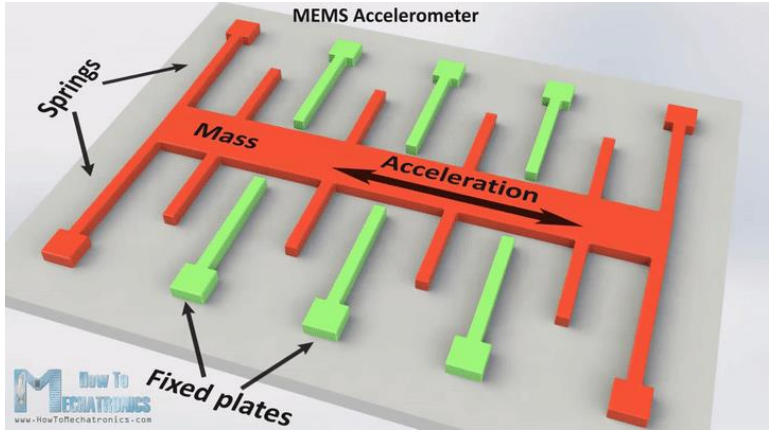
Major drawback of magnetic solutions

- easily disturbed by magnetic objects or other sources
- **not suitable for indoor environments for absolute orientation**

Accelerometer

An accelerometer is a device measuring all external forces applied to it, <u>including gravity.</u>
Conceptually an accelerometer is a *spring-mass-damper* system

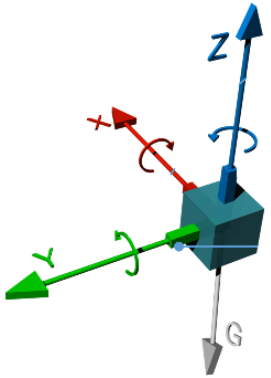
In a mechanical accelerometer, a mass is attached to a spring. Under the influence of an external force, at equilibrium, mass deflection *x* is a measure of the acceleration along spring's axis, accounting for the damping effect (coefficient *c*)

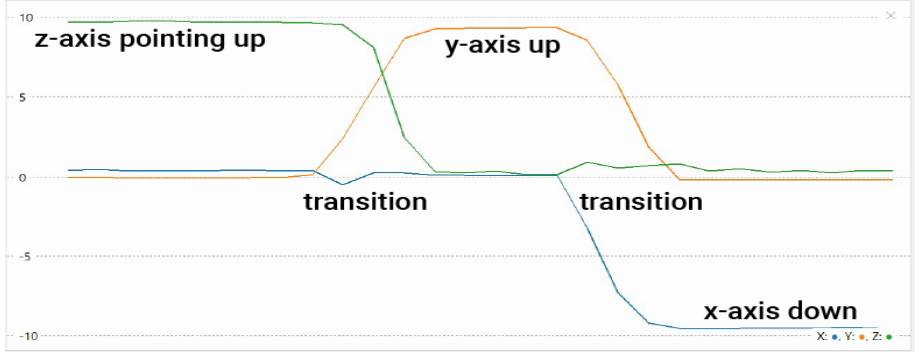


$$F_{applied} = F_{inertial} + F_{damping} + F_{spring} = m\ddot{x} + c\dot{x} + kx$$

$$At\ equilibrium\ (\ddot{x} = 0),\ a_{applied} = \frac{kx}{m}$$

Mounting 3 accelerometers in 3 orthogonal directions, measuring **reaction force** from 100 Hz up to 100 KHz

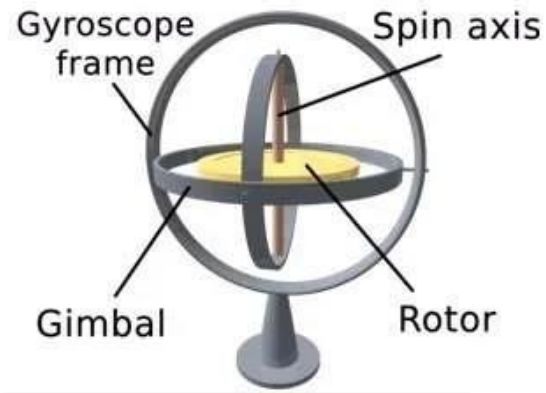


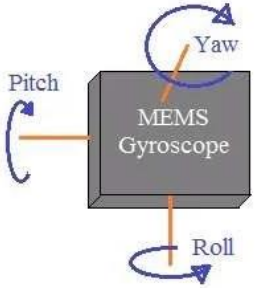


Gyroscope

Gyroscopes are ***heading sensors*** that preserve the orientation in relation to a fixed reference frame, allowing to measure the ***angular velocity*** ω relative to the inertial space

Maintain angular momentum (mechanical gyro). A spinning wheel will resist any change in its angular momentum vector relative to inertial space. Isolating the wheel from vehicle angular movements by means of gimbals and then output the gimbal positions is the idea of a mechanical gyro.



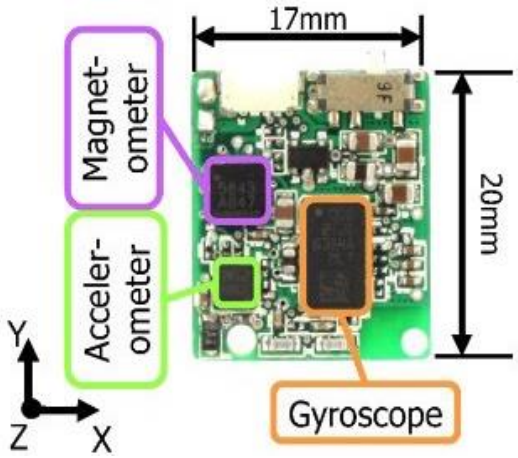


The angular velocity is measured around the spinning axis

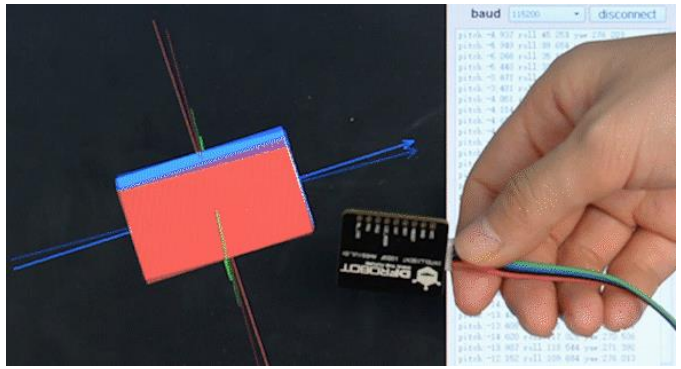
Issue: friction in the bearings of the gyro axis introduce small torques, limiting long-term space stability and introducing small errors over time (e.g., 0.1 degrees / 6 hours for good gyros)

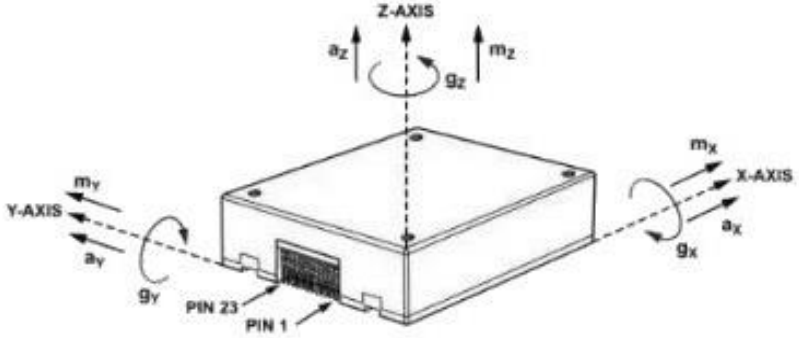
Inertial Measurement Unit (IMU)

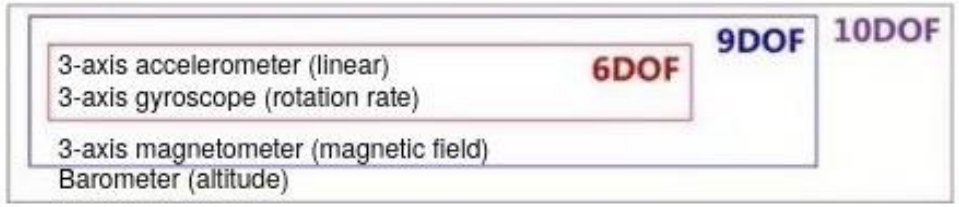
Multiple inertial sensors are often *assembled* together to form an **Inertial Measurement Unit (IMU)**



- Typically the unit has 3 accelerometers and 3 gyroscopes (6-DoF IMU)
- In a Strapdown IMU all inertial sensors are rigidly attached to the unit (no mechanical movement)
- The accelerometers are placed such that their measuring axes are orthogonal to each other.
- The gyroscopes are placed in a similar orthogonal pattern, measuring rotational position in reference to an arbitrarily chosen coordinate system



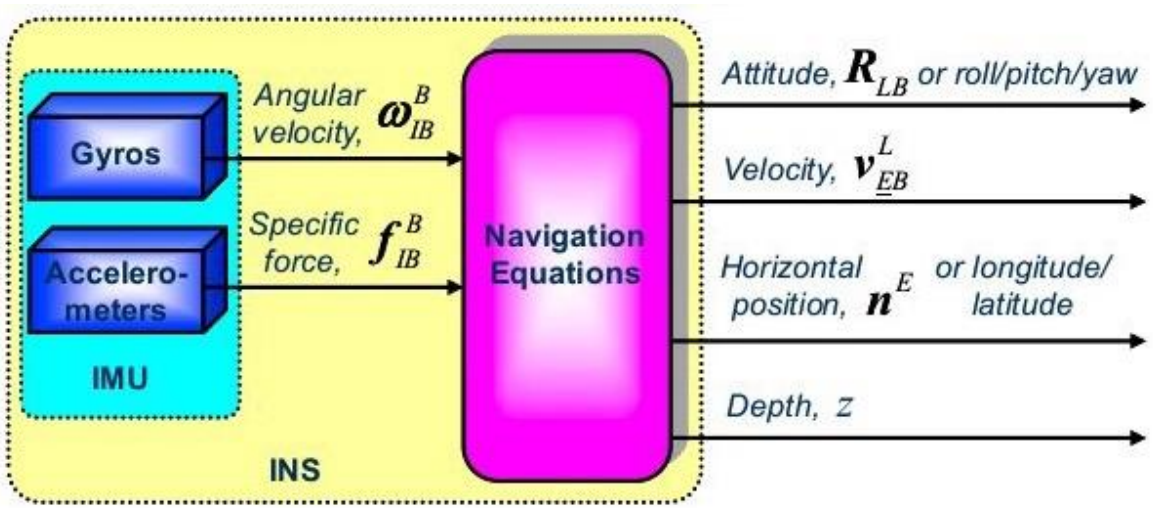




- Optionally, 3 magnetometers / compasses can be also placed (9-DoF) with / without a barometer (10-DoF)

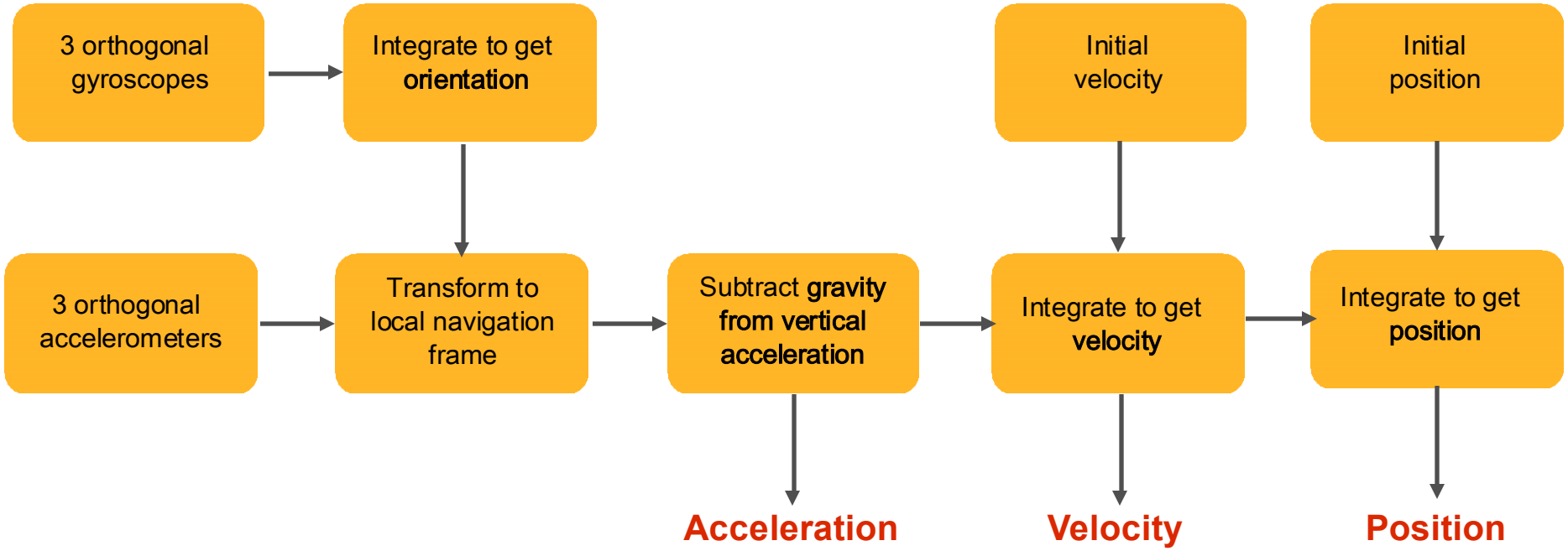
Inertial Measurement Unit (IMU) → Inertial Navigation System (INS)

Inertial Navigation System (INS): combination of an IMU and a computer running dead-reckoning algorithm to continuously calculate position, orientation and velocity of a moving object without the need for external systems.



Due to errors in the gyros and accelerometers and the nature of dead-reckoning algorithm, an INS has **unlimited drift** in position, velocity and likely attitude (some using magnetometers to correct heading).

INS Functional Diagram



Note: The accelerometer will measure all the forces that are applied to robots. Gravity will always be there. Therefore, ***g*** has to be subtracted in order to get the effective acceleration ***a*** that a robot is experiencing. For instance, a planar robot that moves straight on a road with a linearly increasing velocity $v_x = kt$, for what concerns its motion it will experience a constant acceleration $\boldsymbol{a} = (a_x, 0, 0)$.

On the other hand, the measure from the accelerometer will be $\boldsymbol{a} = (a_x, 0, g)$.

INS: Typical Drift Errors

	Accelerometer Bias Error		Horizontal Position Error [m]			
Grade	[mg]		1s	10s	60s	1hr
Navigation	0.025		0.13 mm	12 mm	0.44 m	1.6 km
Tactical	0.3		1.5 mm	150 mm	5.3 m	19 km
Industrial	3		15 mm	1.5 m	53 m	190 km
Automotive	125		620 mm	60 m	2.2 km	7900 km

Accelerometer Misalignment		Horizontal Position Error [m]			
[mg]		1s	10s	60s	1hr
0.05°		4.3 mm	0.43 m	15 m	57 km
0.1°		8.6 mm	0.86 m	31 m	110 km
0.5°		43 mm	4.3 m	150 m	570 km
1°		86 mm	8.6 m	310 m	1100 km

	Gyro Angle Random Walk (ARW)		Horizontal Position Error [m]			
Grade	[deg/√hr]		1s	10s	60s	1hr
Navigation	0.002		0.01 mm	0.1 mm	1.3 mm	620 m
Tactical	0.07		0.1 mm	3.2 mm	46 m	22 km
Industrial	3		10 mm	0.23 m	3.3 m	1500 km
Automotive	5		20 mm	0.45 m	6.6 m	3100 km

Global Positioning Sensors

- **Satellite radio beacons** (GPS)
- **Ground radio beacons** (UWB or WiFi anchors, RFID markers)

Exteroceptive Sensors

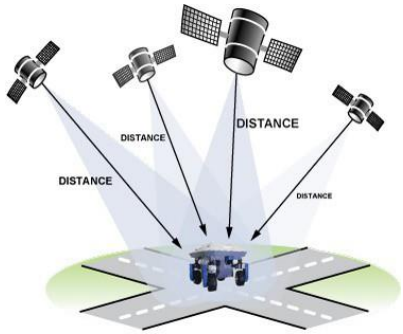
Global Positioning Sensors

Beacon-based positioning / navigation

Active vs. Passive *Landmarks* in the environment

Natural vs. Artificial

Ground vs. Satellite

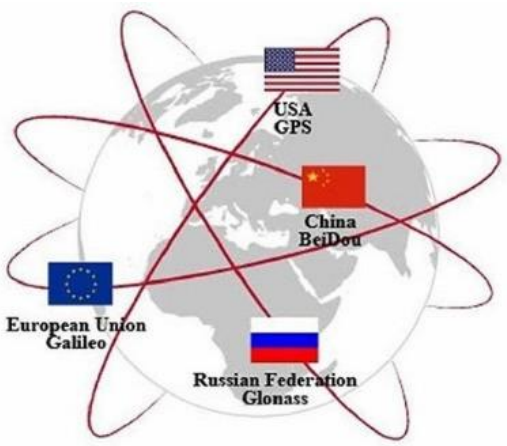


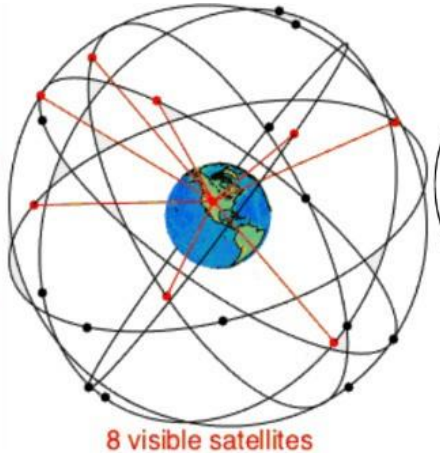
Global Positioning System (GPS)

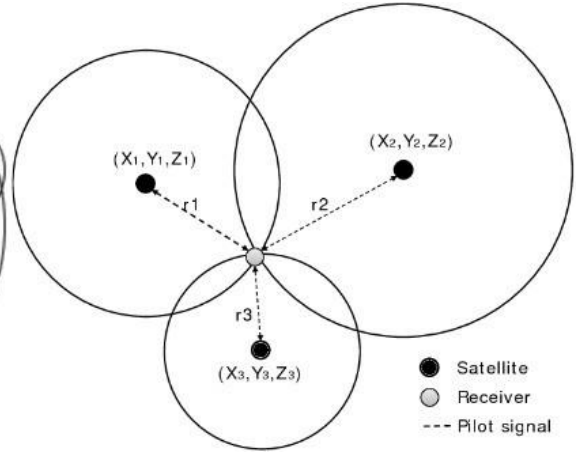
Based on satellites orbiting the earth

Working Principle

- All satellites broadcast in sync their position
- Different TOF due to different satellite distances from the receiver
- Trilateration of distance measures

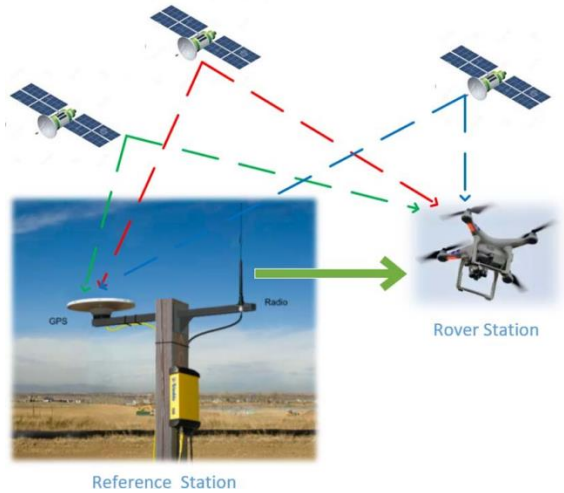






GPS global position measurement: (latitude, longitude)

- Nominal precision: ~10 meters (it can be brought to 2-3m with filtering)
- DGPS (Differential GPS): makes use a reference base station with known location, ~ 0.5 to 3 meters
- RTK (Real-Time Kinematic) GPS: uses measurements of the phase of the signal's carrier wave, and relies on a reference station, providing up to centimeter-level accuracy (~1 to 5 cm)



Range-finder Sensors

- **Ultrasonic sensors:** Time-of-flight of ultrasonics waves
- **Laser range finders:** Time-of-flight of collimated electro-magnetic beams (laser)
- **Radar sensor:** Time-of-flight of radio frequency electromagnetic waves; effective in poor visibility, measures distance and relative velocity via Doppler shift.
- **CCD/CMOS cameras:** Measure gathered intensity of visible light, use disparity or optical flow for distance/range measures
- **RGB-D cameras:** Project a pattern of infrared dots and calculate depth based on distortions
- **Time of flight cameras:** Time-of-flight of infrared collimated (laser/LED) lighting source, matrix of sensors

Exteroceptive sensors

(Active) Range Sensors
Principle - Time of Flight

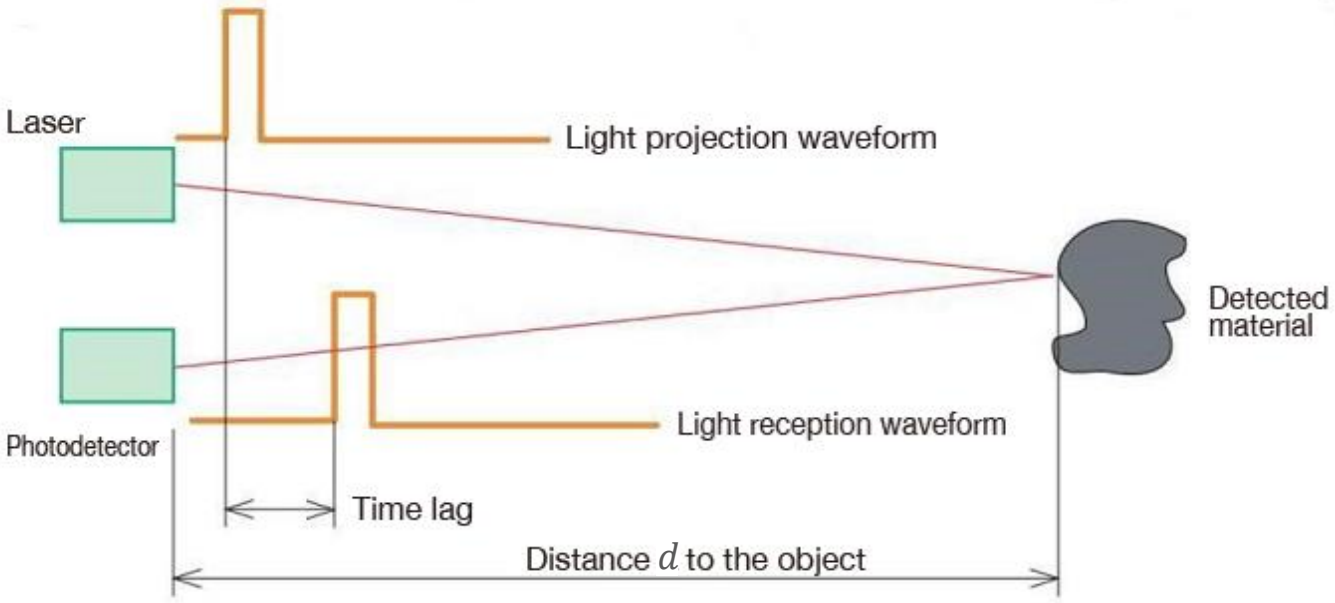
- Range information:
  - key element for localization and (2D/3D) mapping

- Ultrasonic and laser range sensors make use of propagation speed of sound and electromagnetic waves respectively.

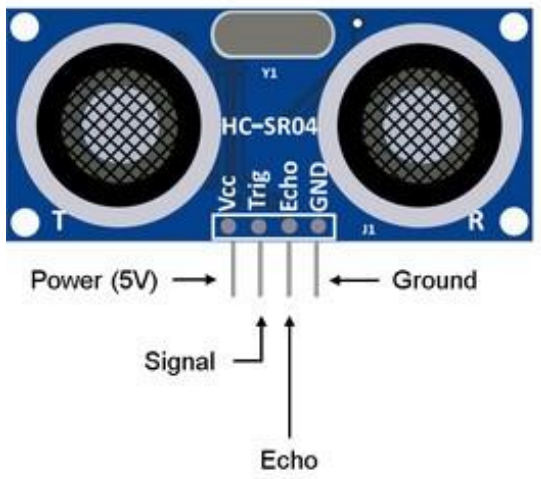
- The traveled distance of a sound or electromagnetic wave is given by

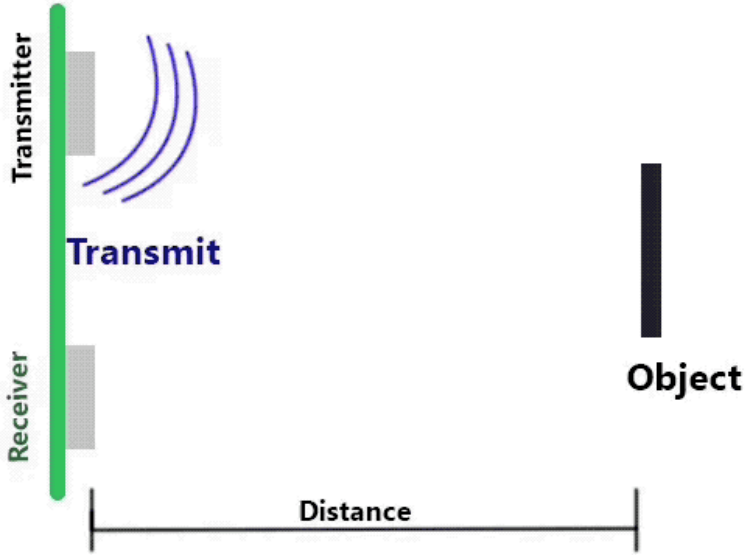
$$d = \frac{c \cdot t}{2}$$

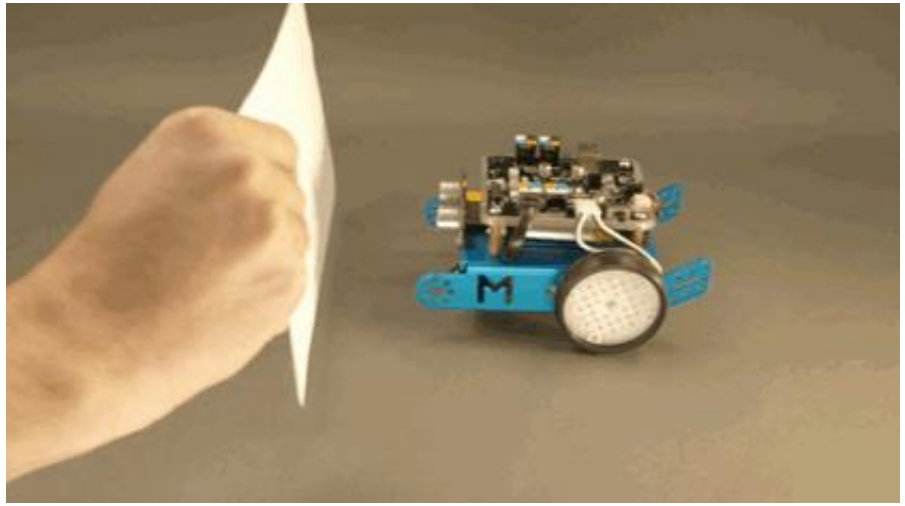
- d = distance traveled
- c = speed of wave propagation
- t = time of flight



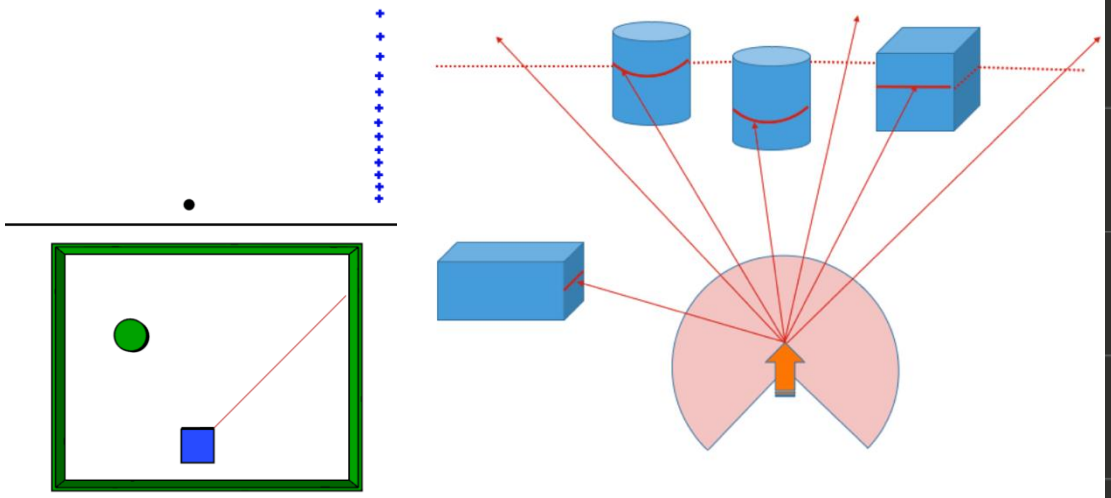
Ultrasonic Sensor

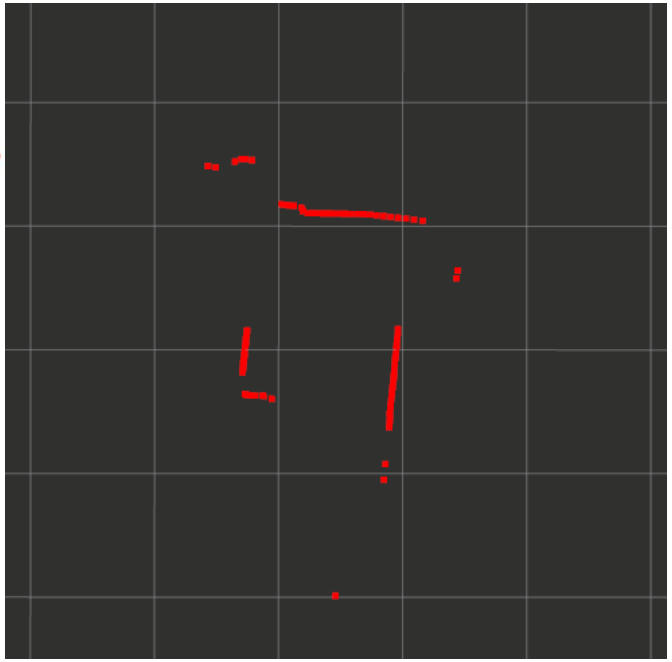




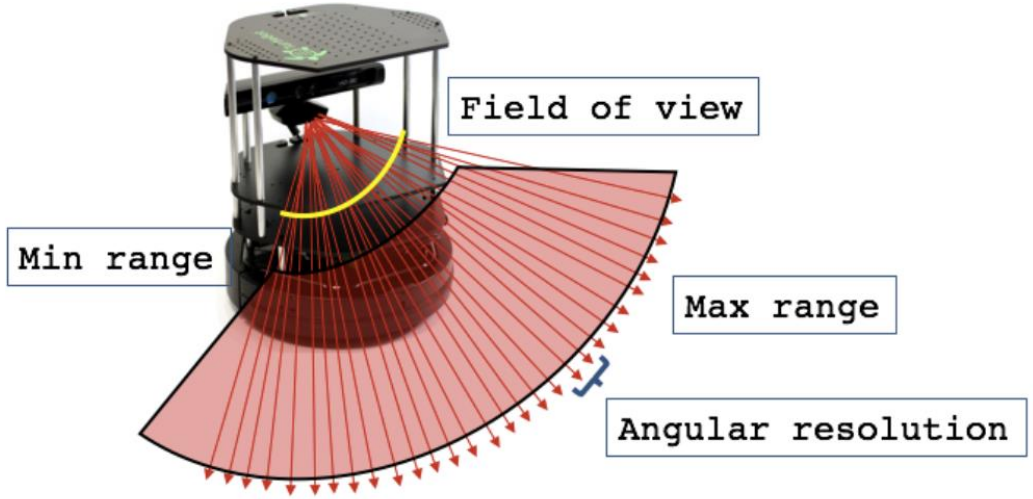


2D Laser Sensor

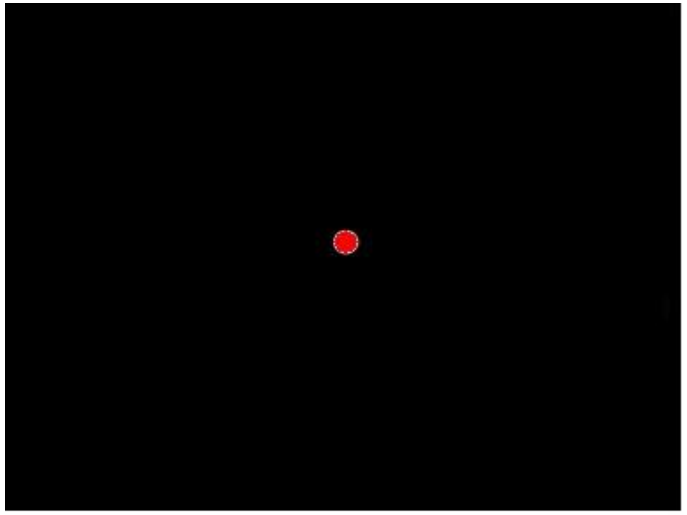




Lidar model	SLAMTEC A1
Ranging radius	0.15m-12m
Ranging frequency	8000Hz
Scanning frequency	5.5Hz-10Hz
Angle resolution	≤1°
Sample range	8k
Communication rate	115200bps
Ranging accuracy	1% of the range (≤3m); 2% of the range(3 - 5m); 2.5% of the range (> 5m)

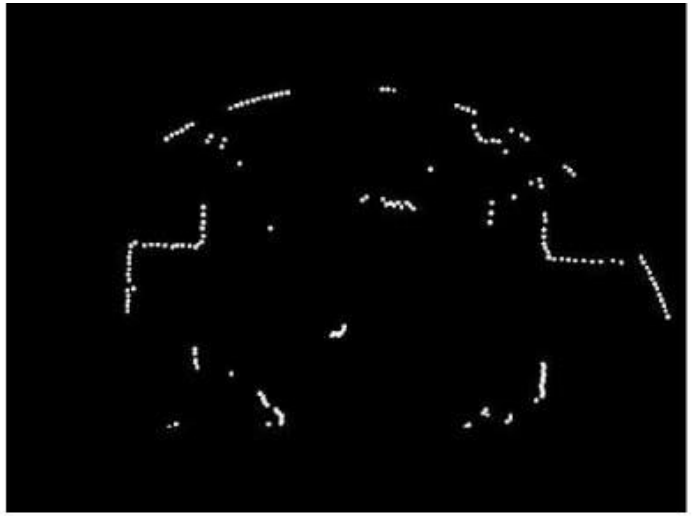


Laser / LiDAR Data



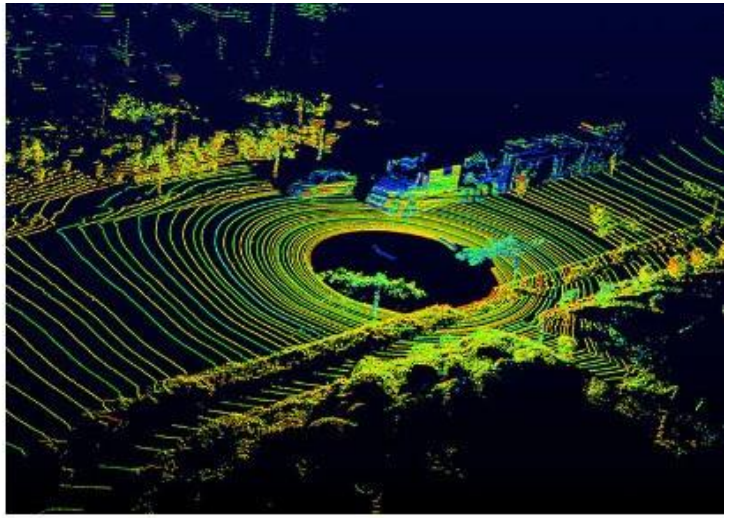
1D

point



2D

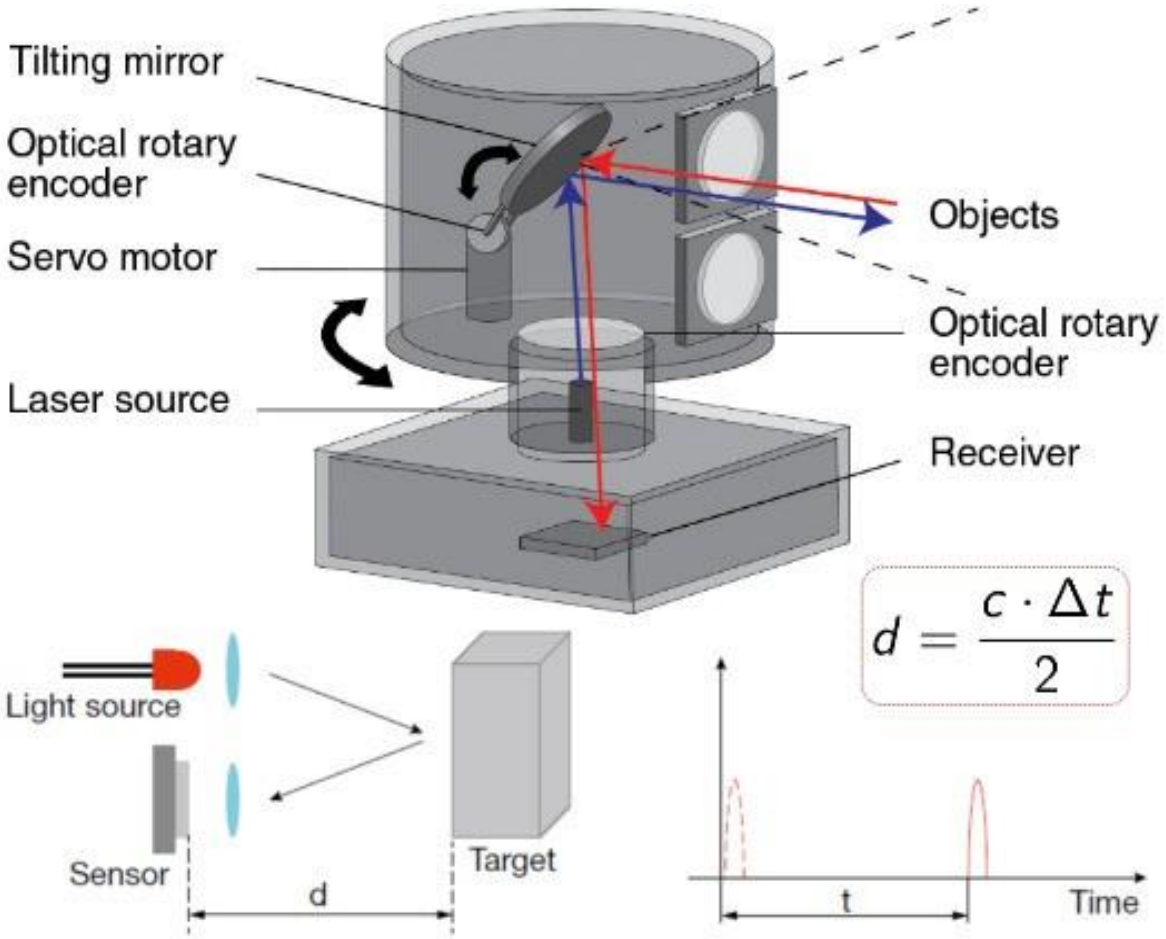
plane
(from a laser scanner)



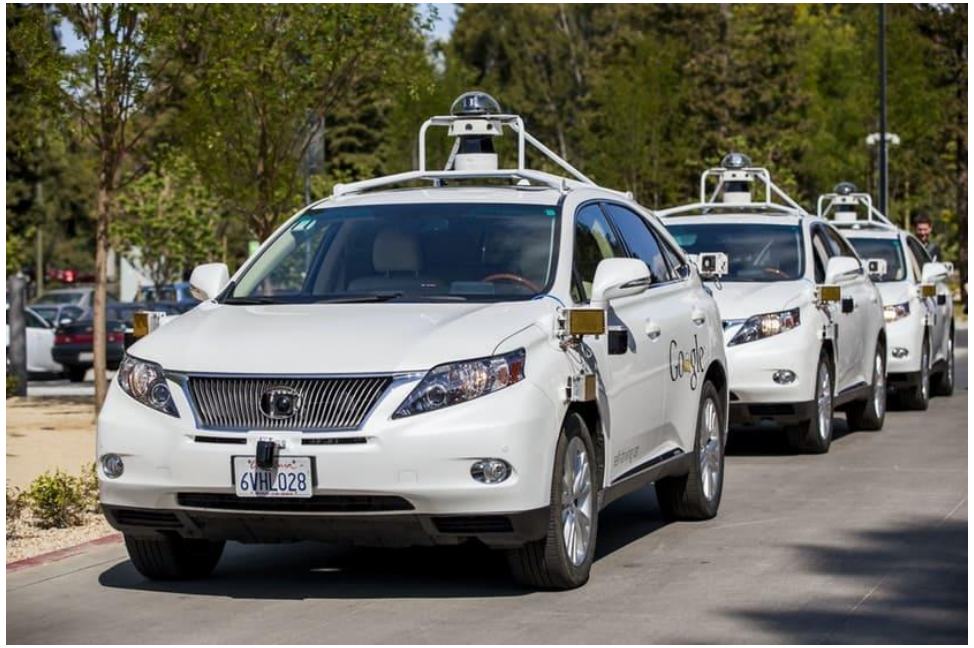
3D

point cloud
(from a LiDAR)

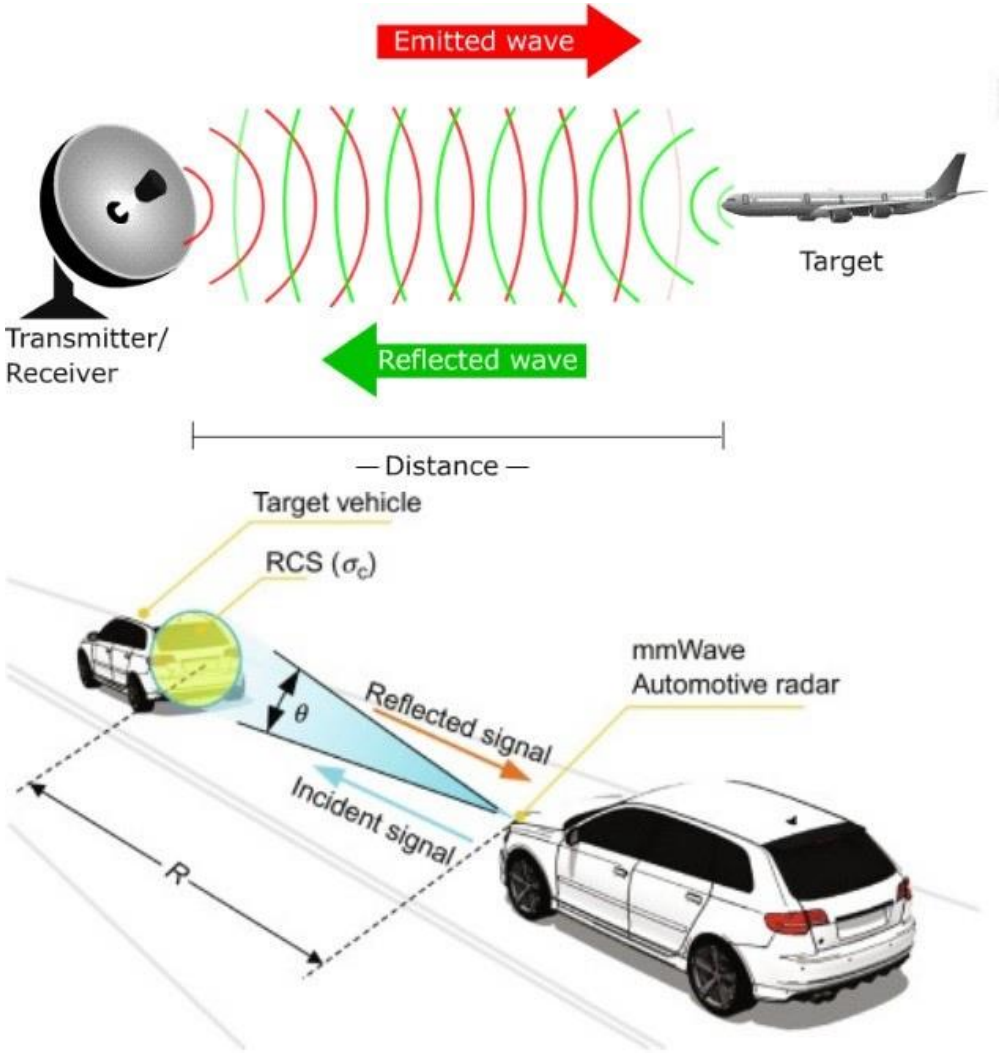
3D Lidar Sensor

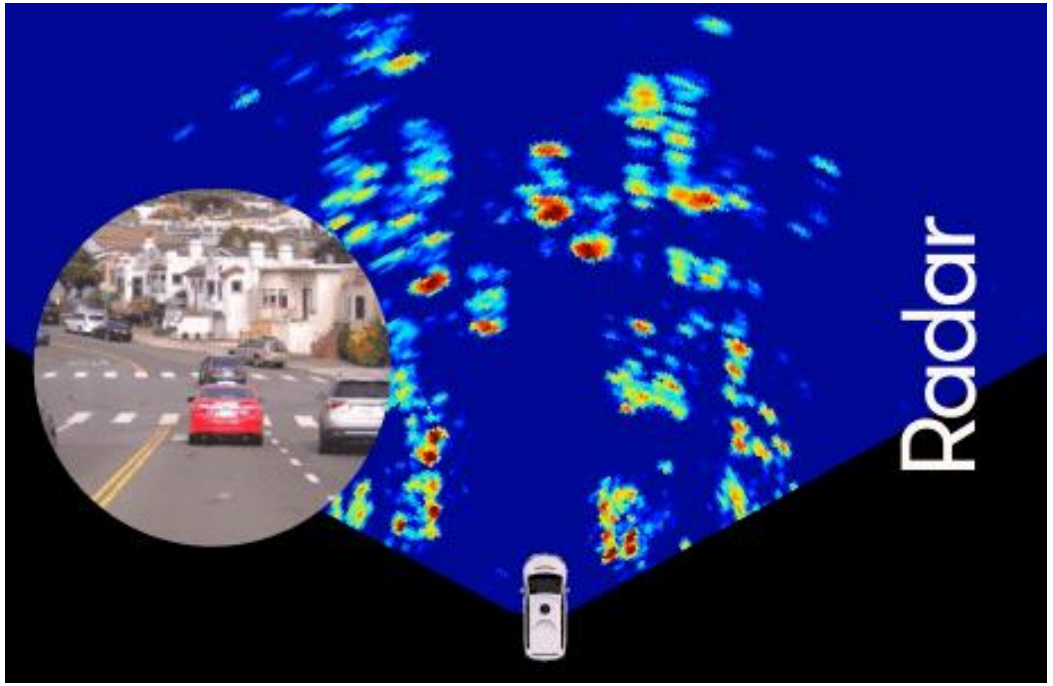


$$d = \frac{c \cdot \Delta t}{2}$$

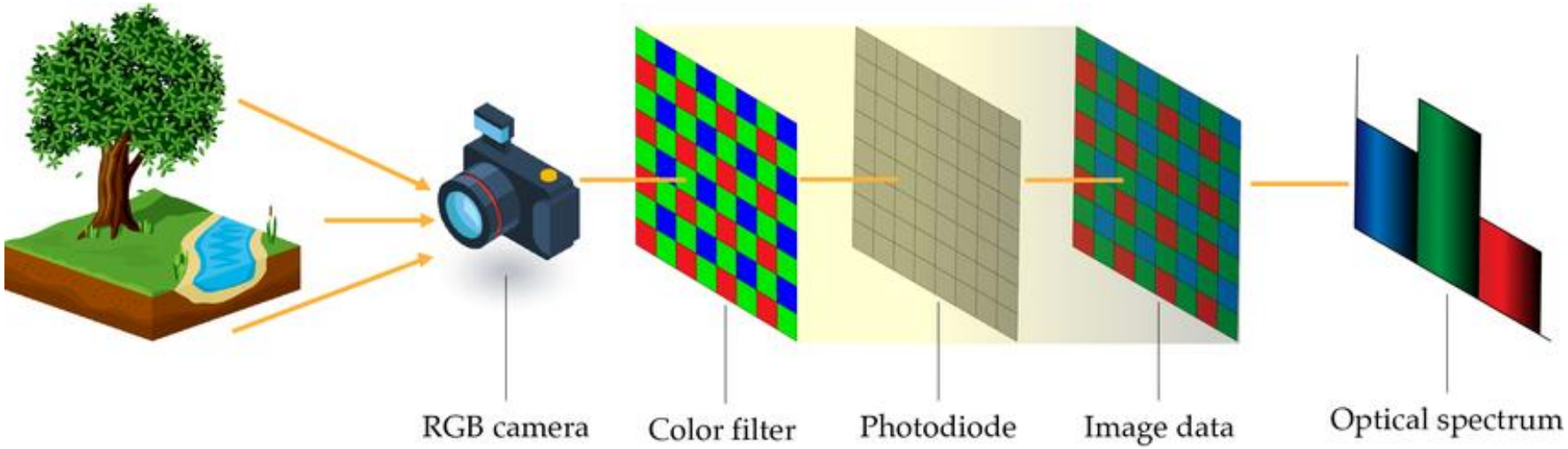


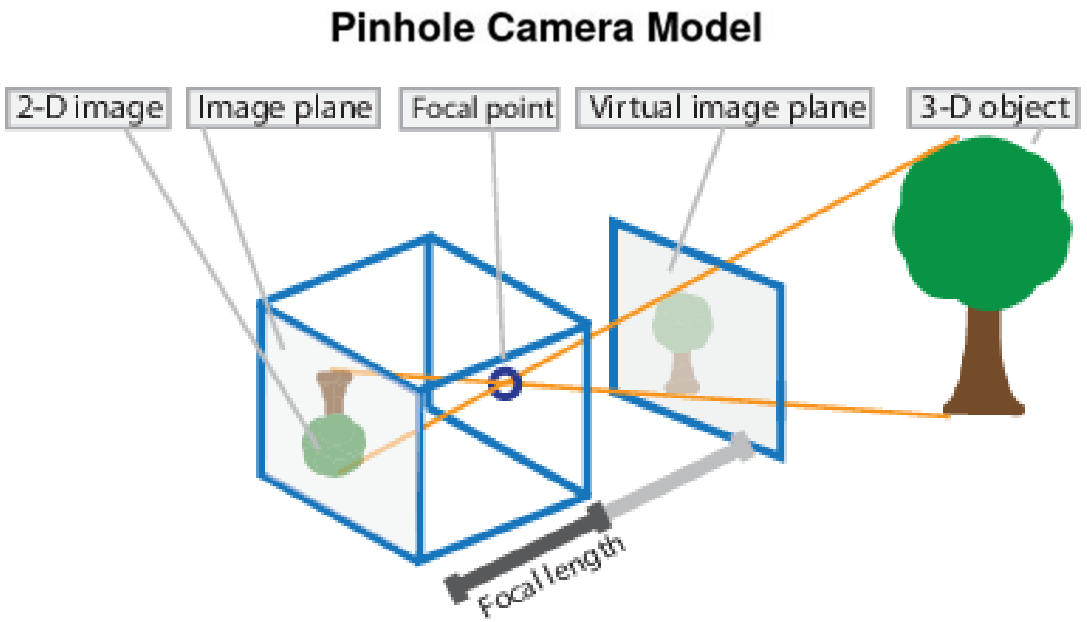
Radar Sensor

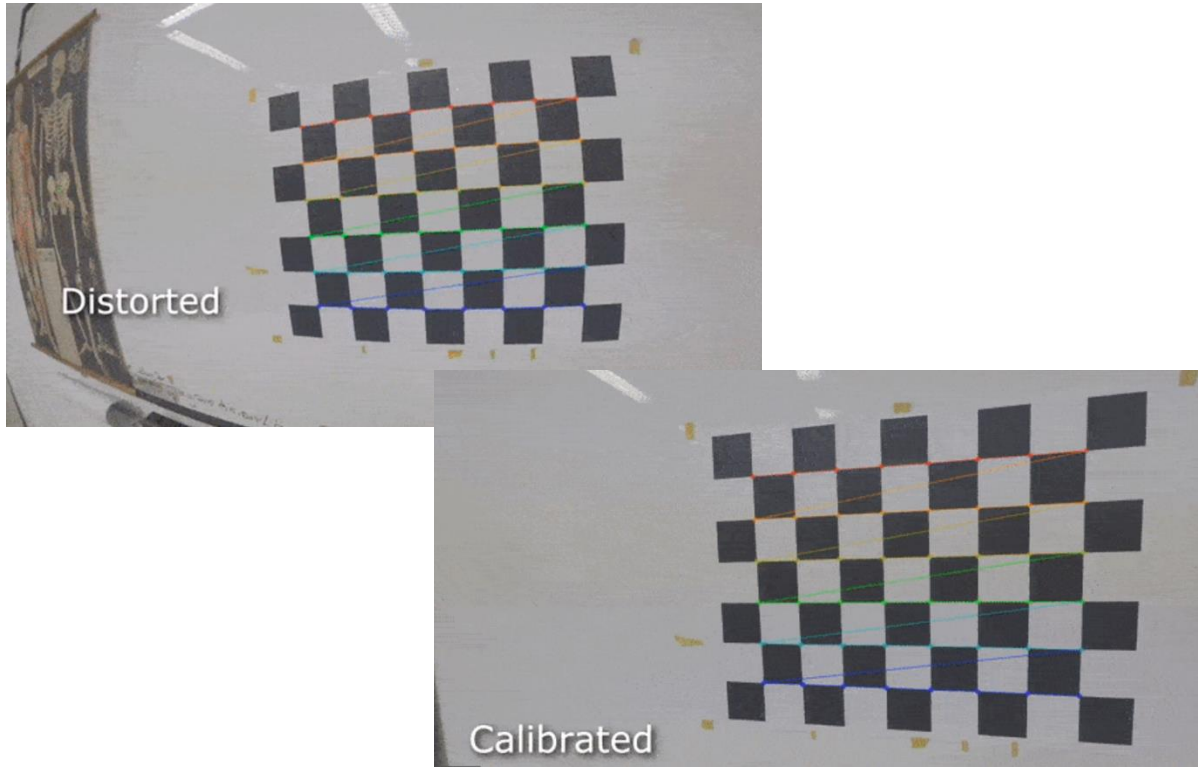




Camera Sensor: RGB



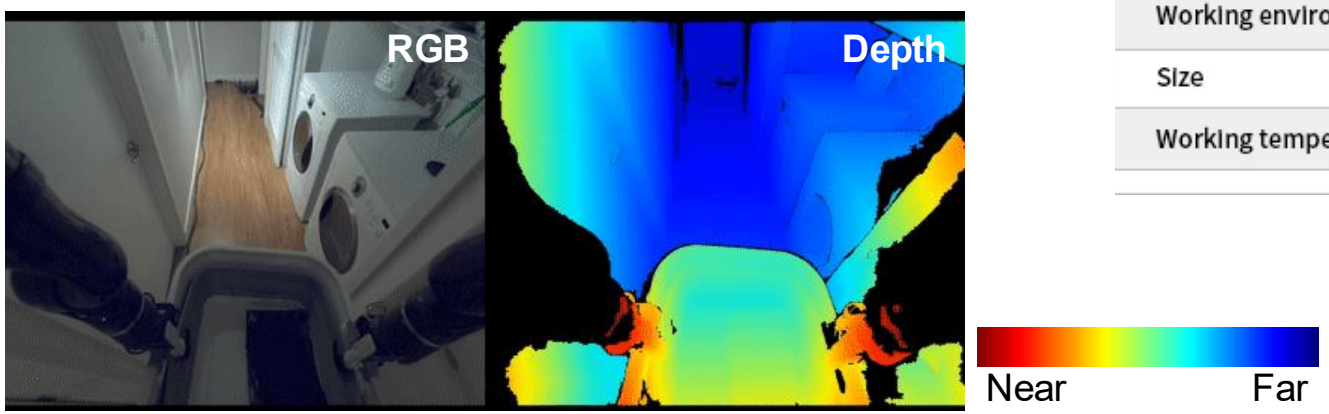




Camera Sensor: RGB-D

RGB-D camera using structured light

- D: Depth
- Projects a pattern of infrared dots and calculates depth based on distortions.

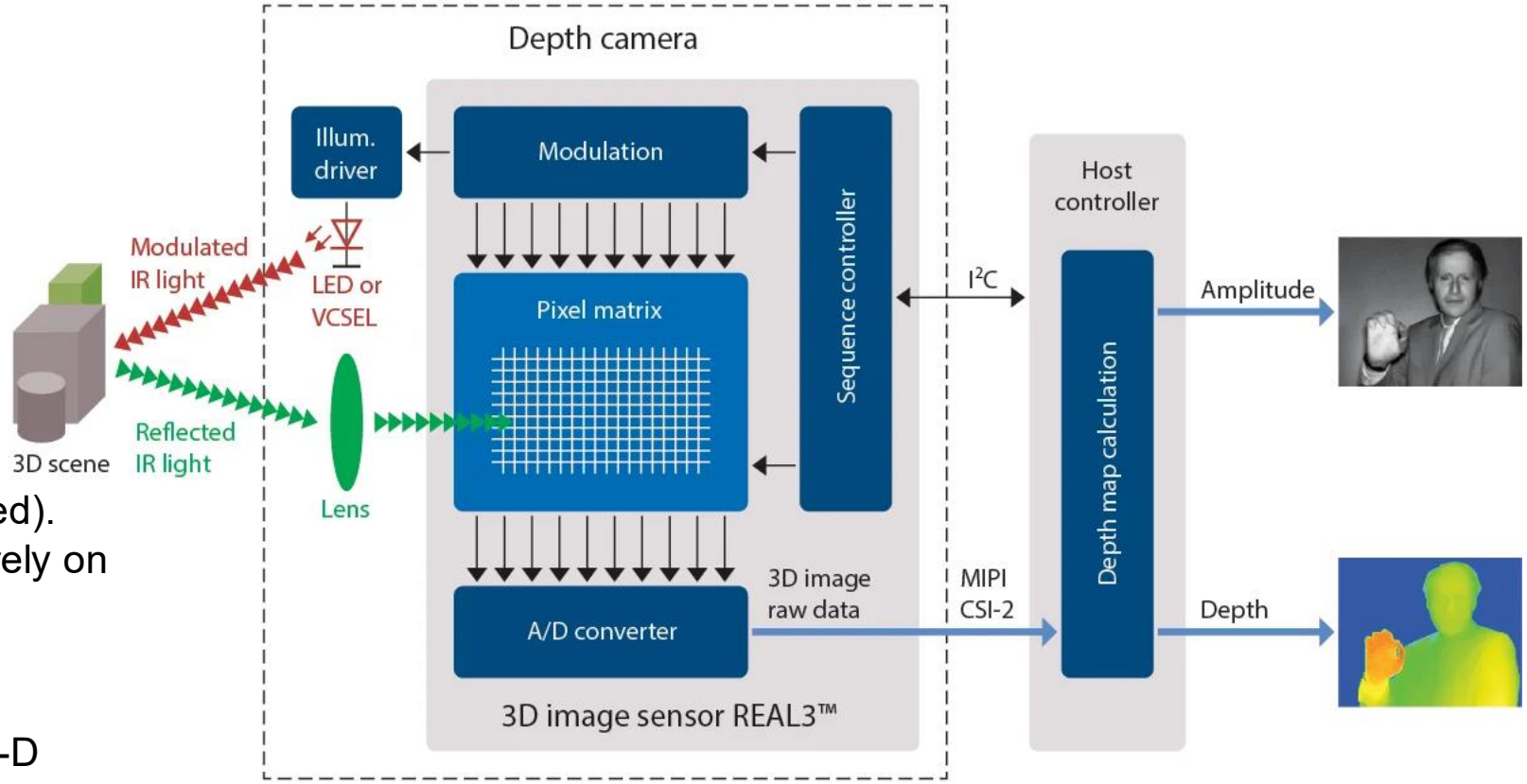


3D technology	ORBBEC® monocular structured light
Accuracy	1m: ±3mm
Field of view	H 66.10°V 40.2°
UVC	support
Resolution @ frame rate	1280×1024@7fps / 640 x480@30fps 320×240@30fps / 160x 120 @30fps
Support operating system	Android/ Linux/ Windows
Data interface	USB2.0
Power consumption	< 2.5W
Security	Class1 Laser
Scope of work	0.6 - 8m
Field of view (FOV)	H58.4° x V 45.7°
Depth processing chip	MX4000
Close protection	support
Resolution @frame rate (RGB mode)	1280 × 720@7fps / 640 × 480@30fps 320 × 240@30fps
Working environment	indoor
Size	164.85 * 48.25 * 40(mm)
Working temperature	10 - 40℃

Camera Sensor: ToF

ToF cameras use infrared light pulses to measure the time it takes for the light to travel to an object and back, determining depth based on time delay.

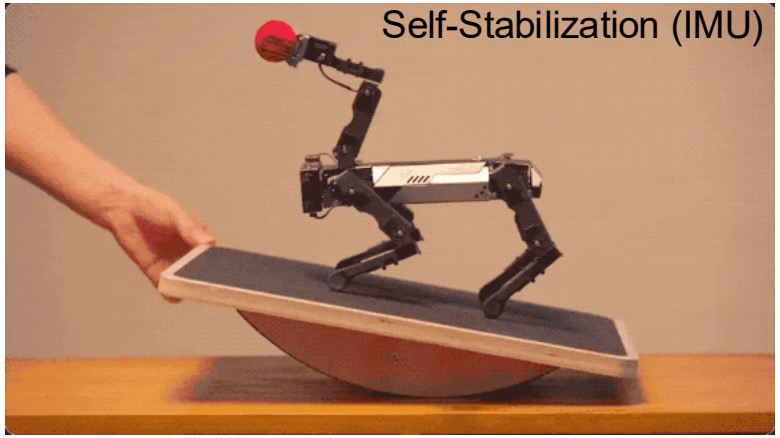
Measures the time of flight of
individual photons



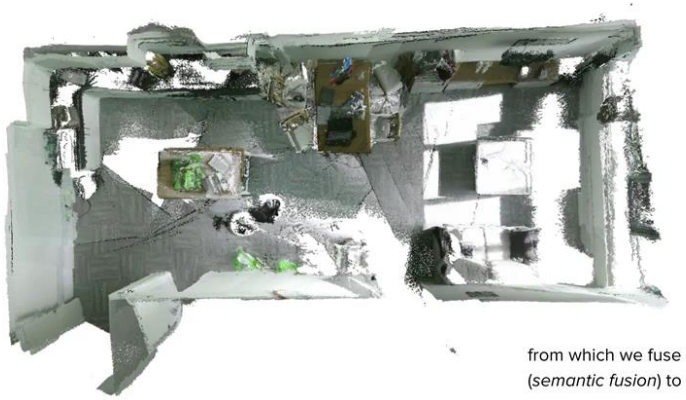
Pros:
- Real-time performance (high frame rates).
- Works in low-light conditions (infrared-based).
- Less interference from textures (does not rely on visible features).

Cons:
- Lower spatial resolution compared to RGB-D cameras.
- Limited outdoor performance due to sunlight interference.

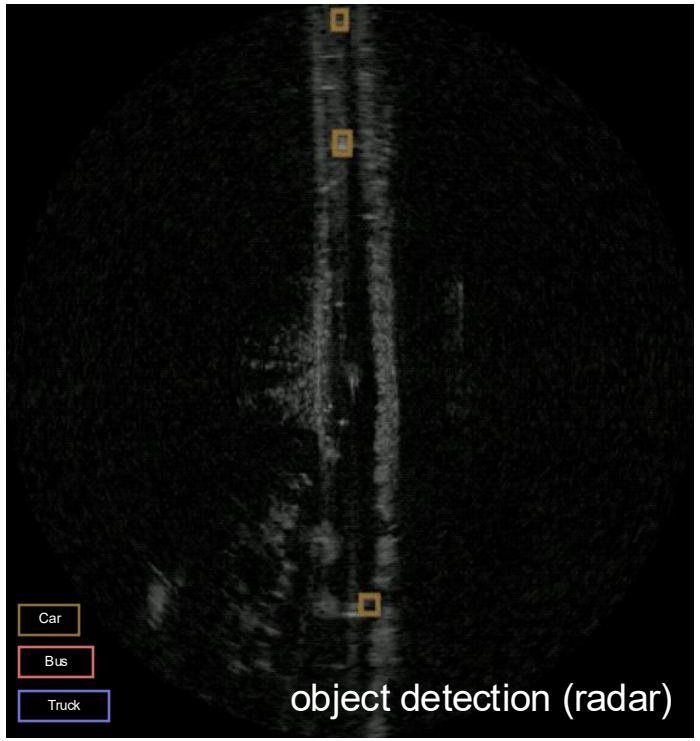
Sensor Data
Exemplar Usages

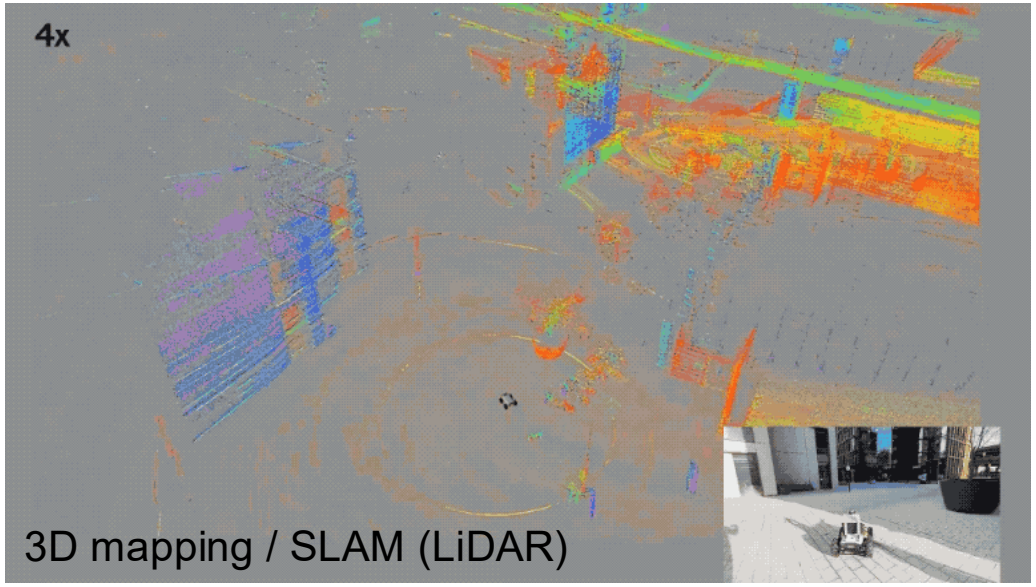


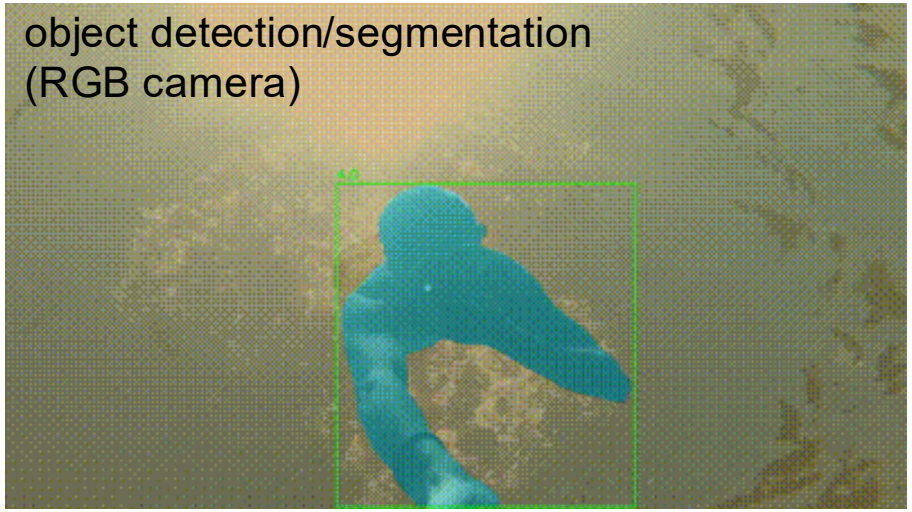
Localisation/Mapping (RGB-D camera)



from which we fuse dense semantic features (*semantic fusion*) to build the global semantic map.





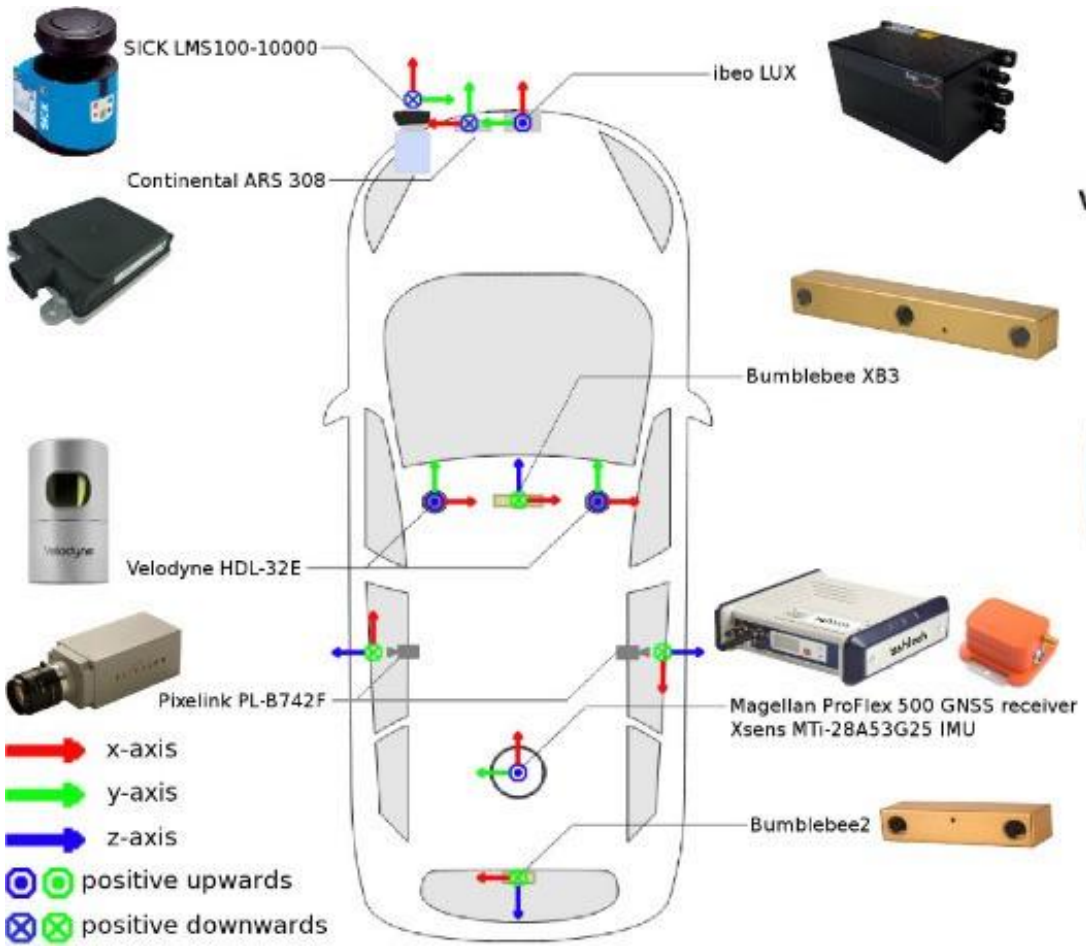


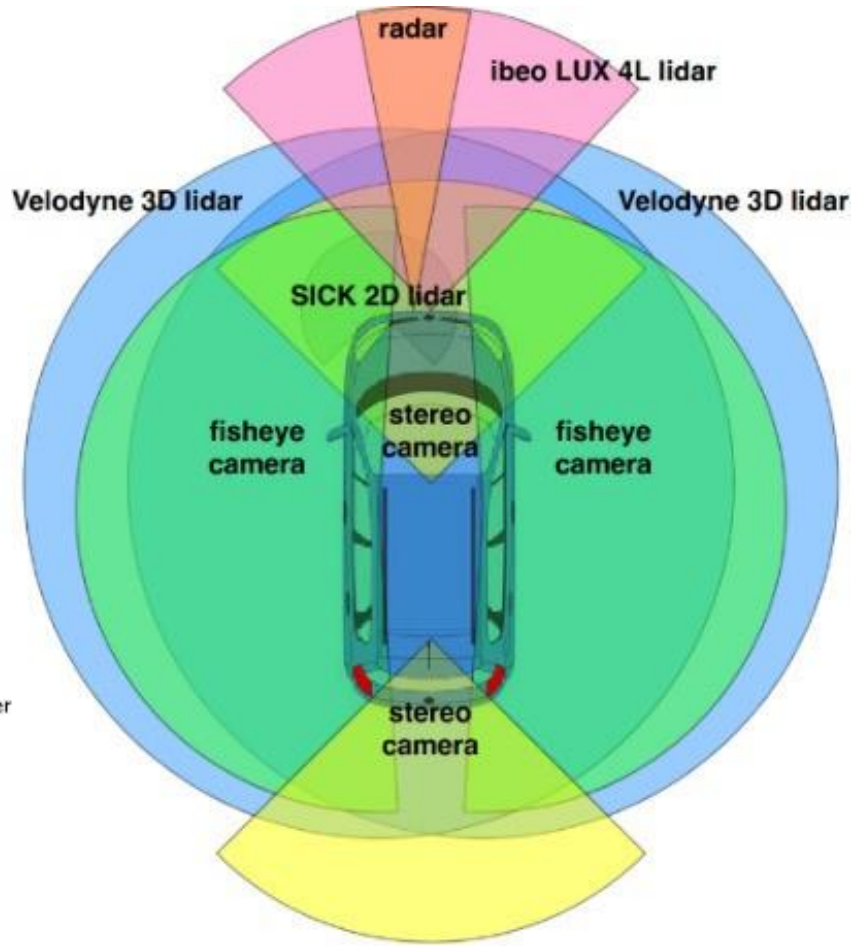
Which Sensors to Use?

Sensors	Pros	Cons
Lidar	-high positioning accuracy -fast data collection -can be used day and night	-high equipment cost -high computational cost -ineffective during rain/etc.
Sonar	-almost all material types -near object detection -unaffected by the weather	-low positioning accuracy -slow data collection -sensitive to temperature
Radar	-reliable detection -unaffected by the weather	-low positioning accuracy -slow data collection
Camera	-low equipment cost -providing intuitive images	-low positioning accuracy -affected by lighting

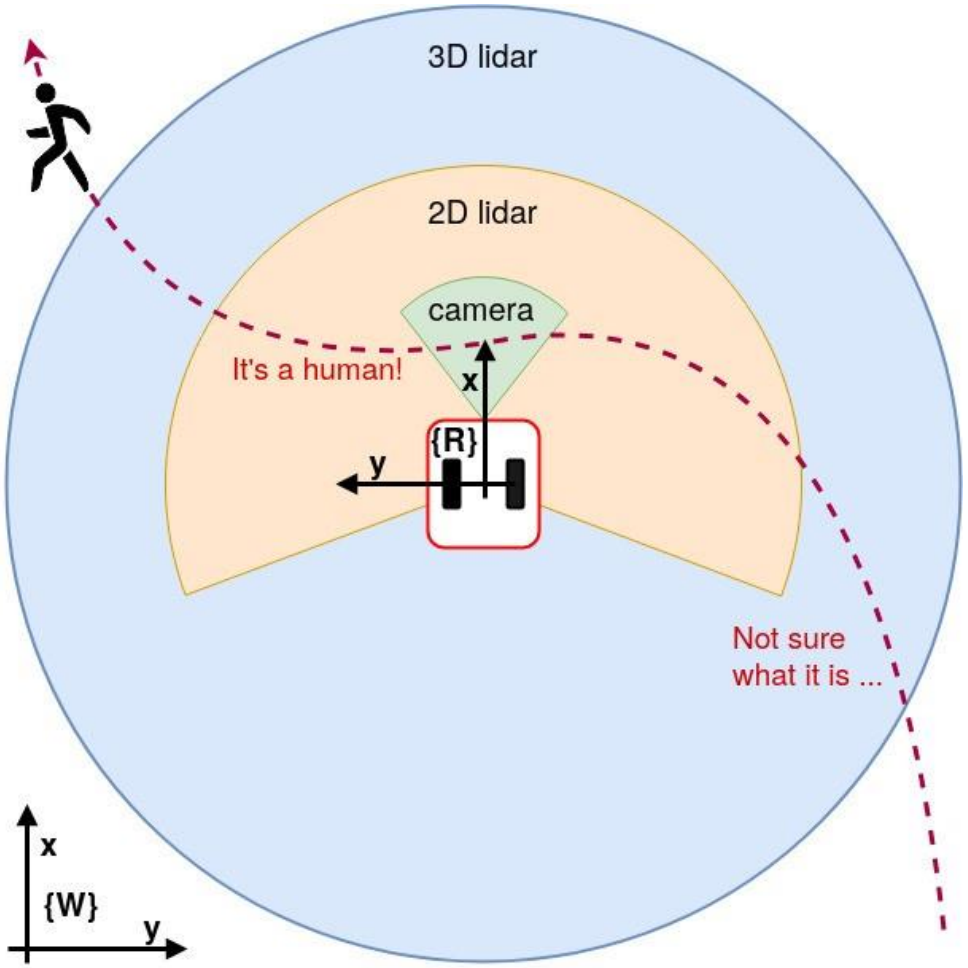
- There is no almighty and perfect sensor, they all have limitations and edge cases.
- Because different sensors have different physical properties and each category has its own pros and cons.

Which Sensors to Use?



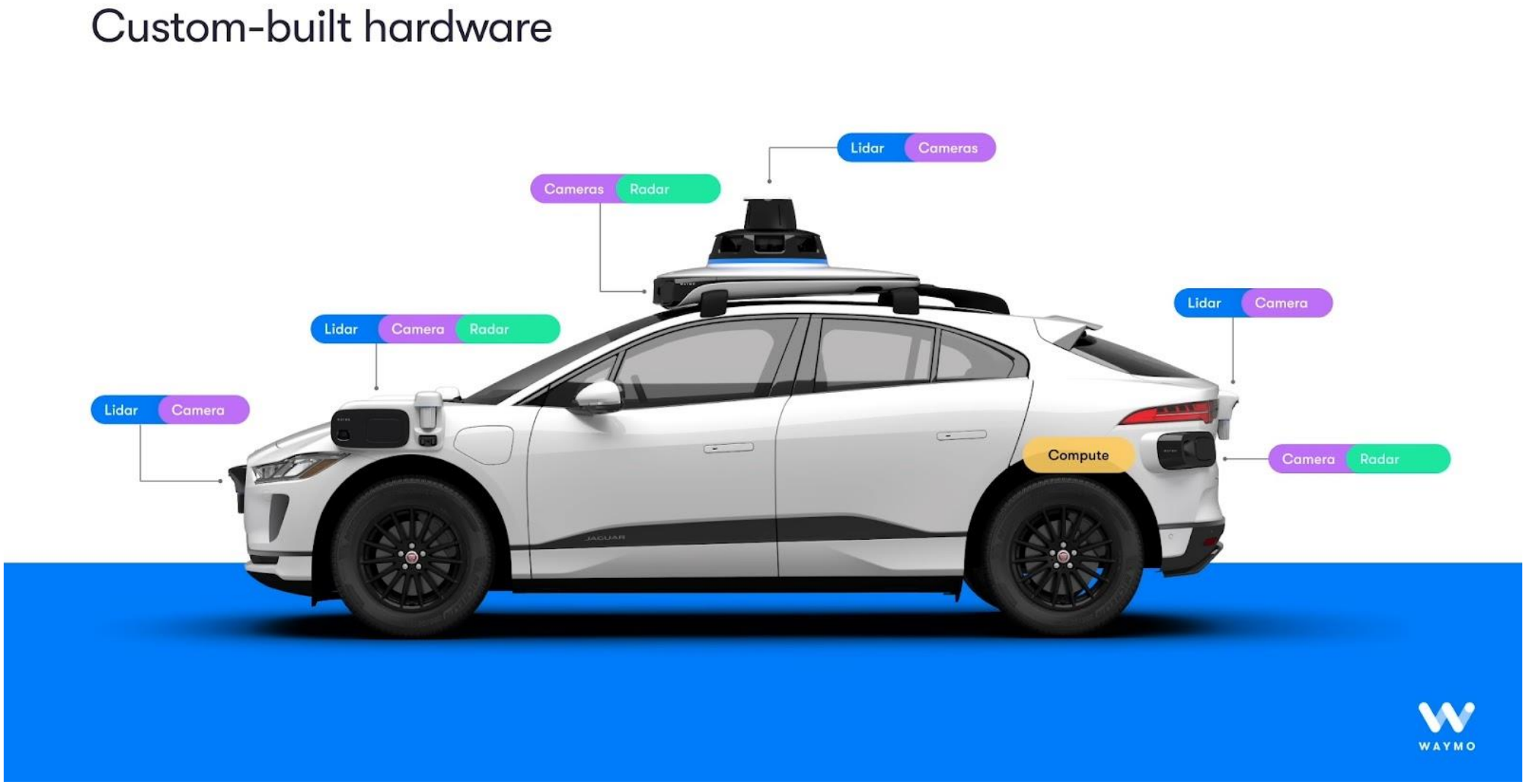


Which Sensors to Use?



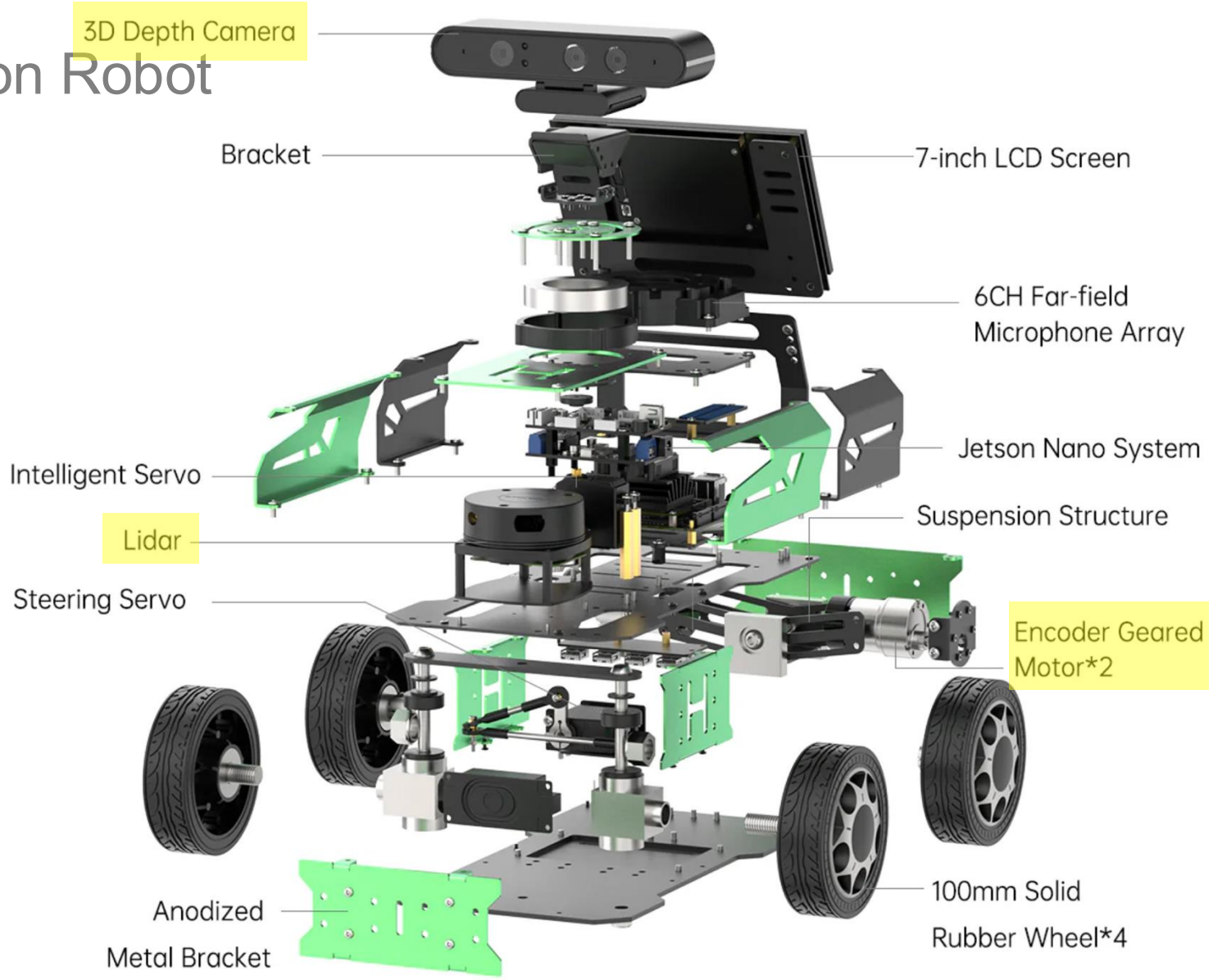
Sensors Used

Example: Self-Driving Cars



Sensors Used

Example: Simulation Robot



Sensor Models

What is a Sensor Model?

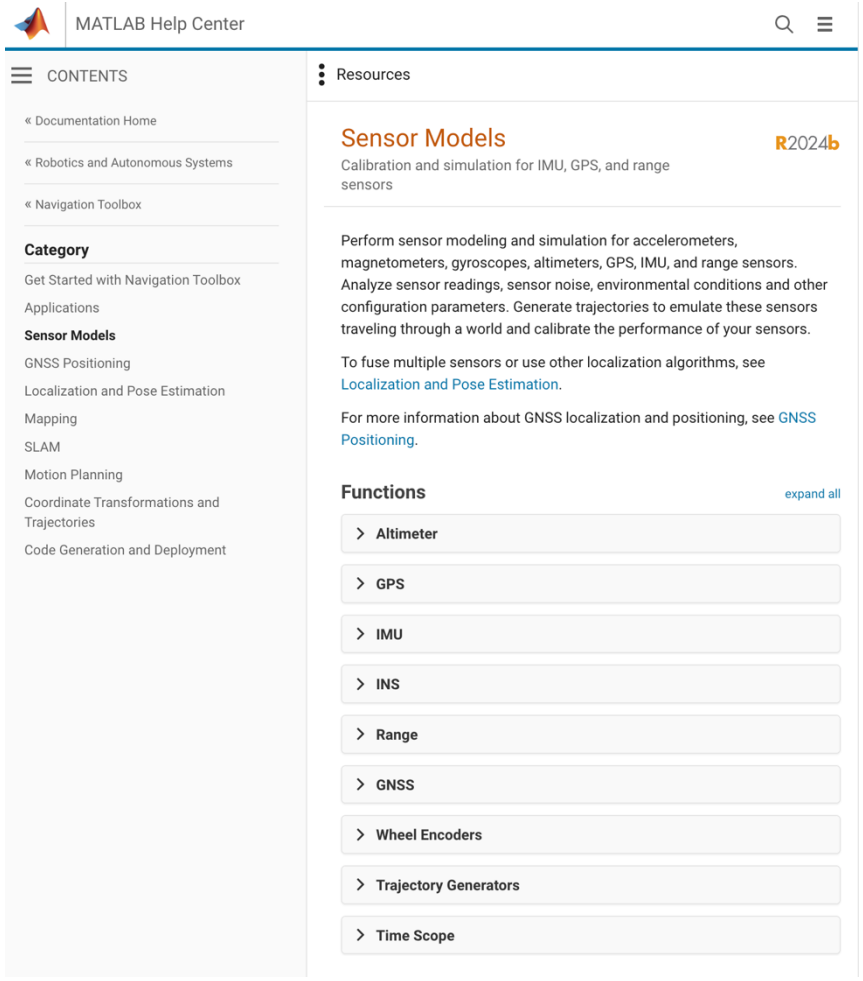
A sensor model is a **mathematical / probabilistic representation** of how a sensor perceives the environment.

It defines the relationship between:

- **actual world / robot state** (e.g., position, object location)
- **sensor measurement** (e.g., GPS reading, image pixel, LiDAR range)

Typically includes:

- noise characteristics (random, systematic)
- measurement uncertainty
- sensor limitations (range, field of view, resolution)



Why Sensor Models Matter

- **State Estimation**: Algorithms (e.g., Kalman Filter, SLAM) require sensor models to fuse data.
- **Sensor Fusion**: Multi-sensor fusion (e.g., IMU + GPS + camera + LiDAR) only works if each model is defined.
- **Simulation and Testing**: Realistic models allow for reliable simulation (ROS Gazebo, Matlab).
- **Error Handling**: Knowing model limitations helps in detecting faults or anomalies.
- **Planning and Control**: Sensor reliability affects decision-making and safety.

In Short: Without a good sensor model, your robot is just guessing.

IMU Sensor Model

Gyroscope Model: gyroscope measures instantaneous angular velocity of {I} relative to {W} expressed in the IMU coordinate frame {I}.

$$_{\mathtt{I}}\tilde{\boldsymbol{\omega}}_{\mathtt{WI}} = {}_{\mathtt{I}}\boldsymbol{\omega}_{\mathtt{WI}} + \mathbf{b}^g + \boldsymbol{\eta}^g$$

measured angular velocity true angular velocity

- gyroscope bias $\mathbf{b}^g$
- measurement noise $\boldsymbol{\eta}^g$: unpredictable variations due to electronic noise or vibrations.

What is bias:

- a constant shift in the readings due to manufacturing imperfections, temperature effects, or sensor drift.

Accelerometer Model: accelerometer measures linear acceleration in three orthogonal axes (x,y,z) with respect to the IMU frame {I}, where the measurement is consistently influenced by gravity.

$$_{\mathtt{I}}\tilde{\mathbf{a}} = \mathbf{R}_{\mathtt{WI}}^T({}_{\mathtt{W}}\mathbf{a} - {}_{\mathtt{W}}\mathbf{g}) + \mathbf{b}^a + \boldsymbol{\eta}^a$$

measured acceleration true acceleration

- ${}_{\mathtt{W}}\mathbf{g} \doteq [0, 0, -g]^T$ the gravity vector, and g is its magnitude (=9.81 m/s^2).
- accelerometer bias $\mathbf{b}^a$
- measurement noise $\boldsymbol{\eta}^a$

GPS Sensor Model

GPS: provides absolute position as geodetic GPS coordinates (latitude, longitude) in earth frame

For sensor model (used for sensor fusion), GPS data should be in a same coordinate system with the robot:

- transform from (latitude, longitude) to local Cartesian coordinates (x, y) in the robot's world frame
- then the measurement model is $\boldsymbol{p}_m = \boldsymbol{p} + \boldsymbol{n}_m$

measured position true position

$$z_k = \begin{bmatrix} z_{x_k} \\ z_{y_k} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} + W_k$$

Refer to ROS navsat_transform_node in robot_localization package or geonav_transform package for info.

https://docs.ros.org/melodic/api/robot_localization/html/navsat_transform_node.html

https://wiki.ros.org/geonav_transform

Laser Scanner / LiDAR Sensor Model

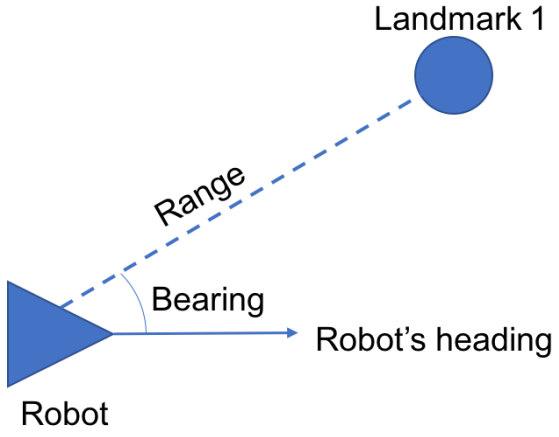
Feature/Landmark/Point based measurement models:
measure range and bearing, like laser scanner / LiDAR

$$r = \| \boldsymbol{p}_f - \boldsymbol{p}_r \| + n_d$$

measured range (distance)

feature position

sensor position

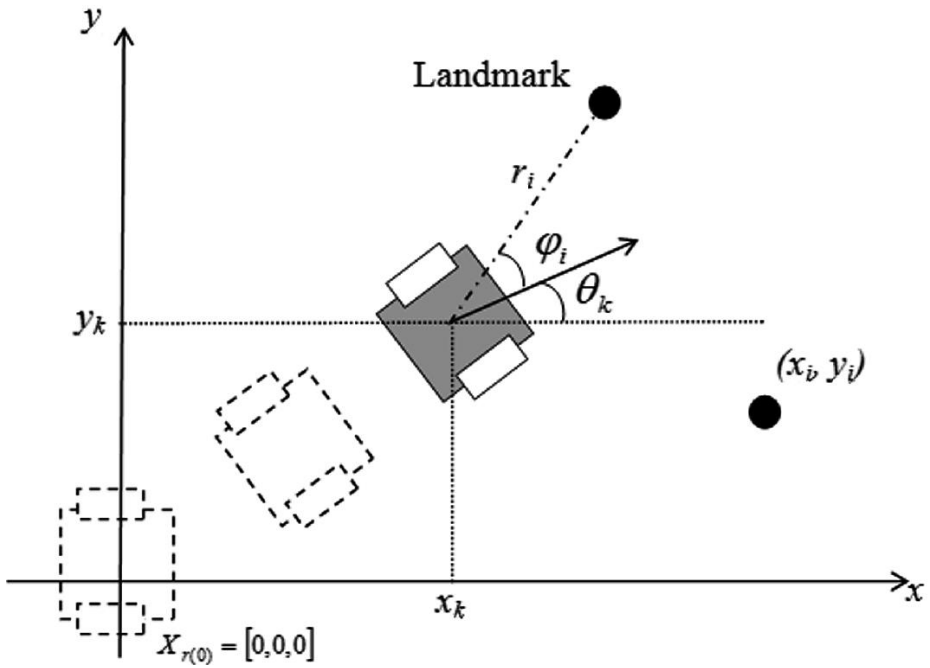


Range-bearing $z_t^i = (r_t^i, \phi_t^i)^T$

Robot's pose $(x, y, \theta)^T$

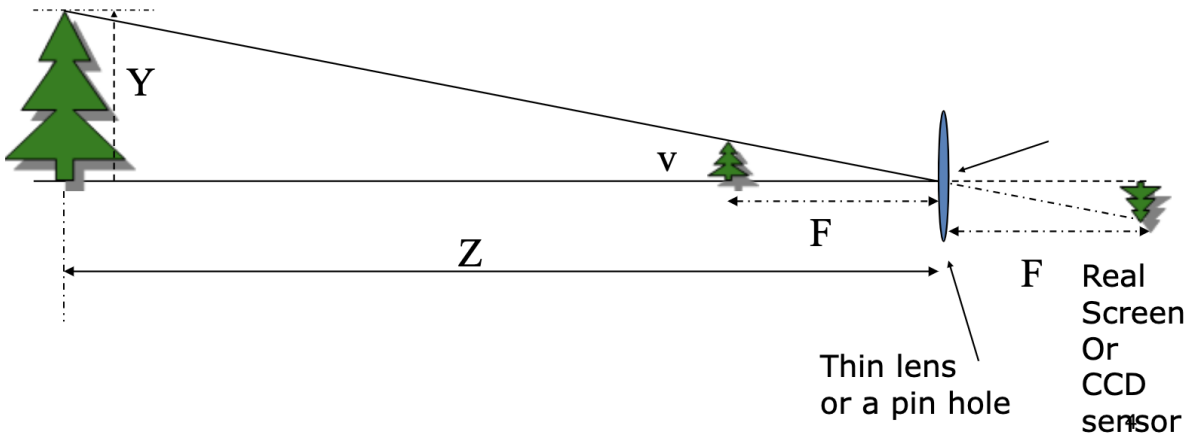
Observation of feature j at location $(m_{j,x}, m_{j,y})^T$

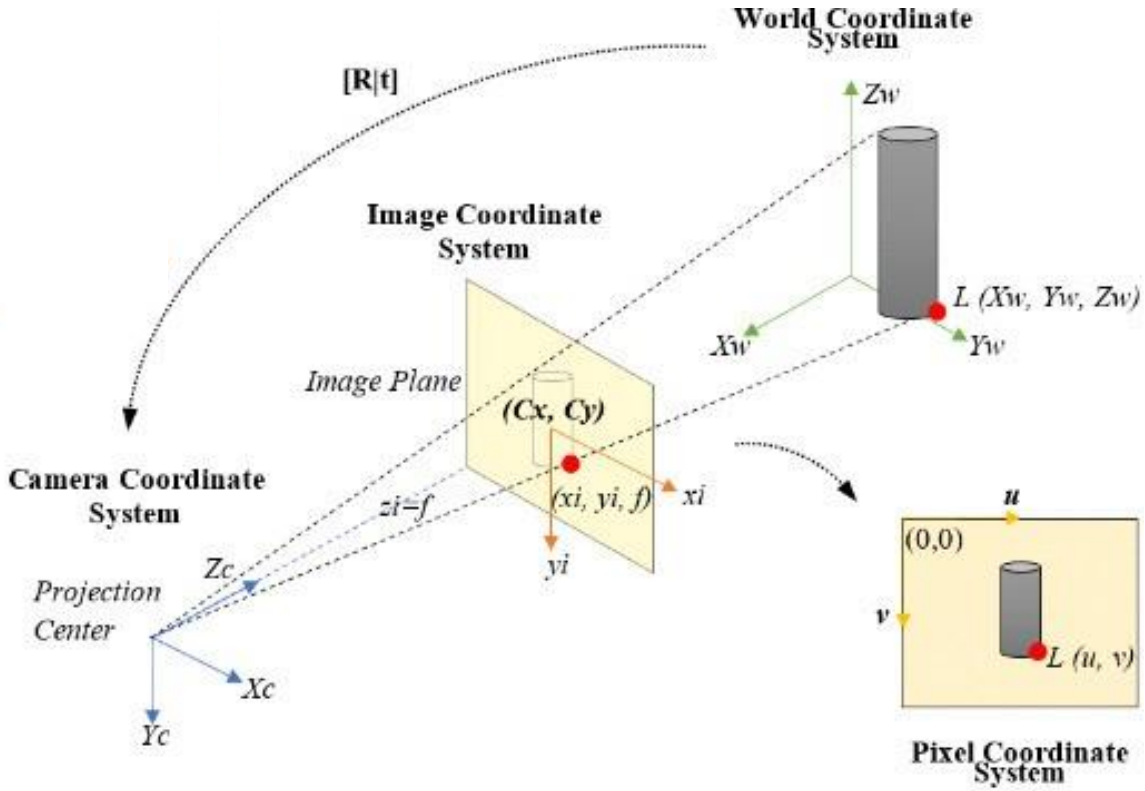
$$\begin{pmatrix} r_t^i \\ \phi_t^i \end{pmatrix} = \begin{pmatrix} \sqrt{(m_{j,x} - x)^2 + (m_{j,y} - y)^2} \\ \operatorname{atan2}(m_{j,y} - y, m_{j,x} - x) - \theta \end{pmatrix} + Q_t$$



Camera Sensor Model

- Captures 2D image projections of 3D world
- Uses perspective projection model
- Affected by lens distortion and lighting conditions





orientation as 3x3 rotation matrix

position as 3x1 vector

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

2D Image Coordinates

Intrinsic properties (Optical Centre, scaling)

Extrinsic properties (Camera Rotation and translation)

3D World Coordinates

measured pixel location
s: scale factor

intrinsic matrix **K**

camera pose in the world frame

Summary

- **Sensors**
  - **Proprioceptive sensors**
    - Motor Encoders (Lecture 4)
    - IMU / INS
      - Magnetometers
      - Accelerometer
      - Gyroscope
  - **Exteroceptive sensors**
    - GPS
    - Ultrasonic
    - Laser and LiDAR
    - Radar
    - Camera: RGB, RGB-D, ToF
- **Sensor models:** widely used sensors

Questions?